

Climate Change Governance: Science, Data & Models

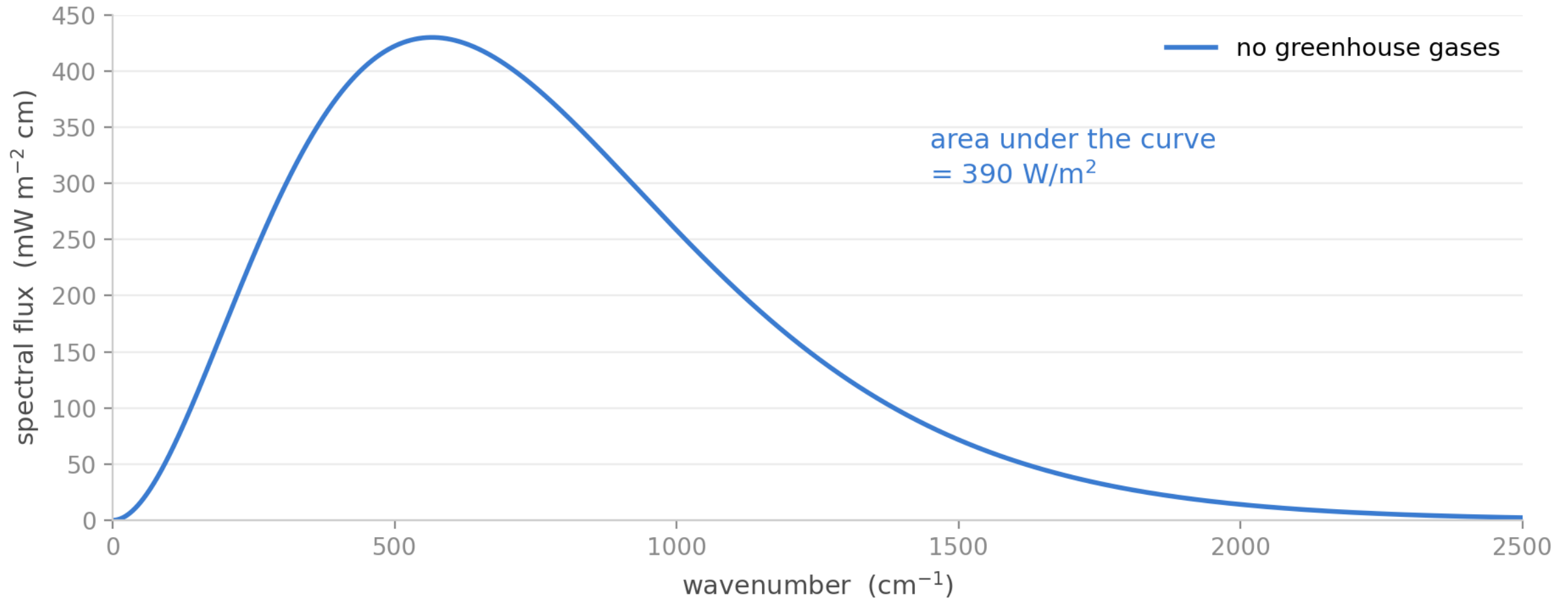


Lecture 3: Observations

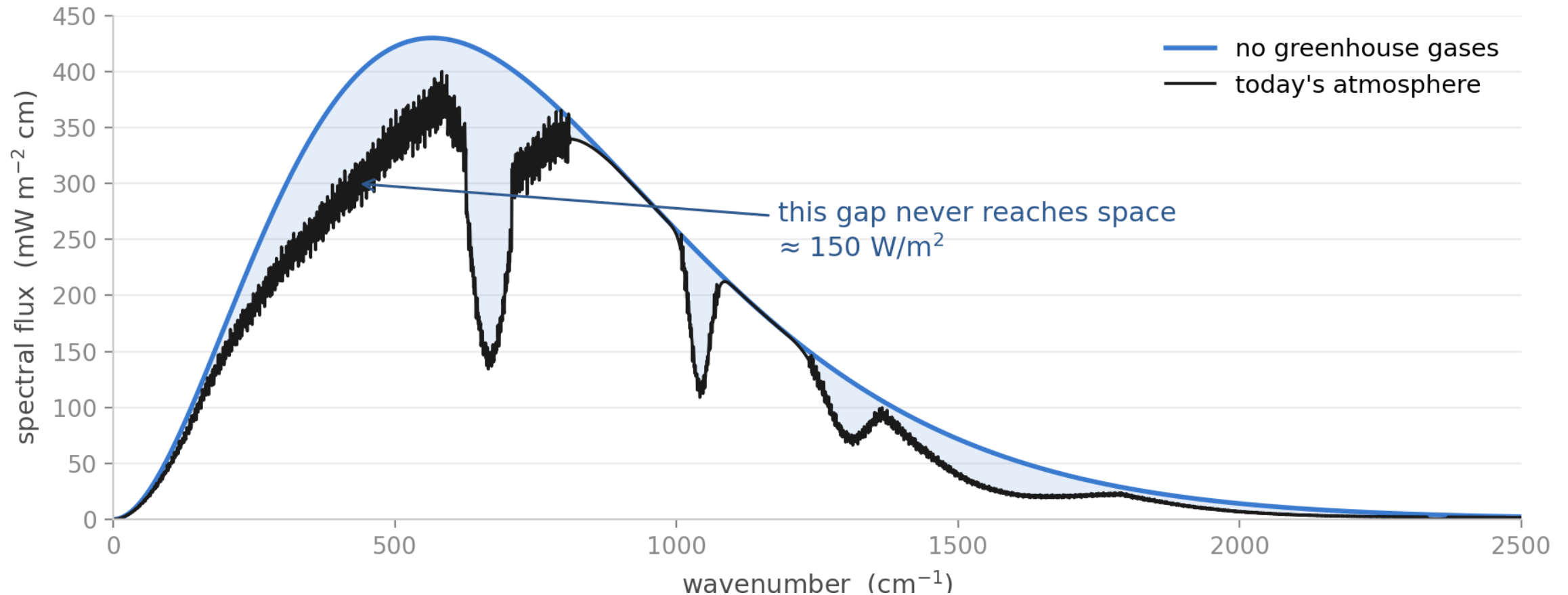
ENGG562/ENV562 – Fall 2026

Muhammad Awais

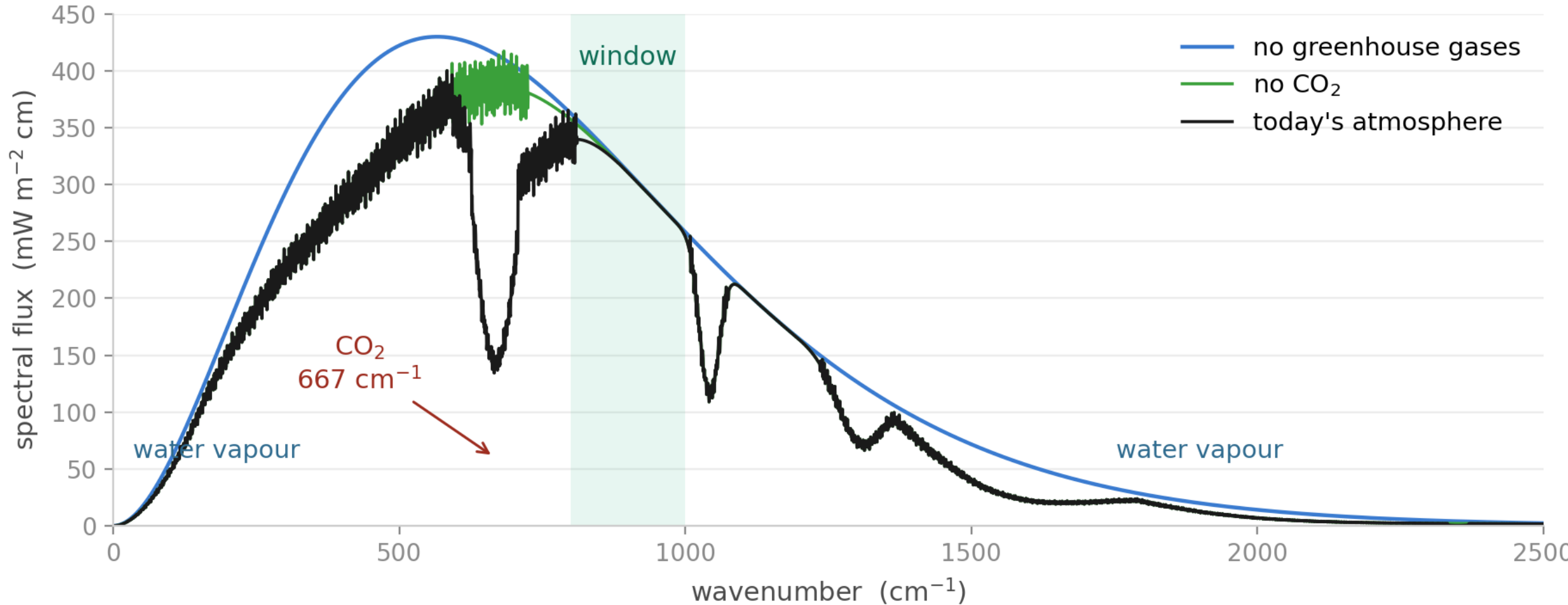
Radiative Forcing – Black Body



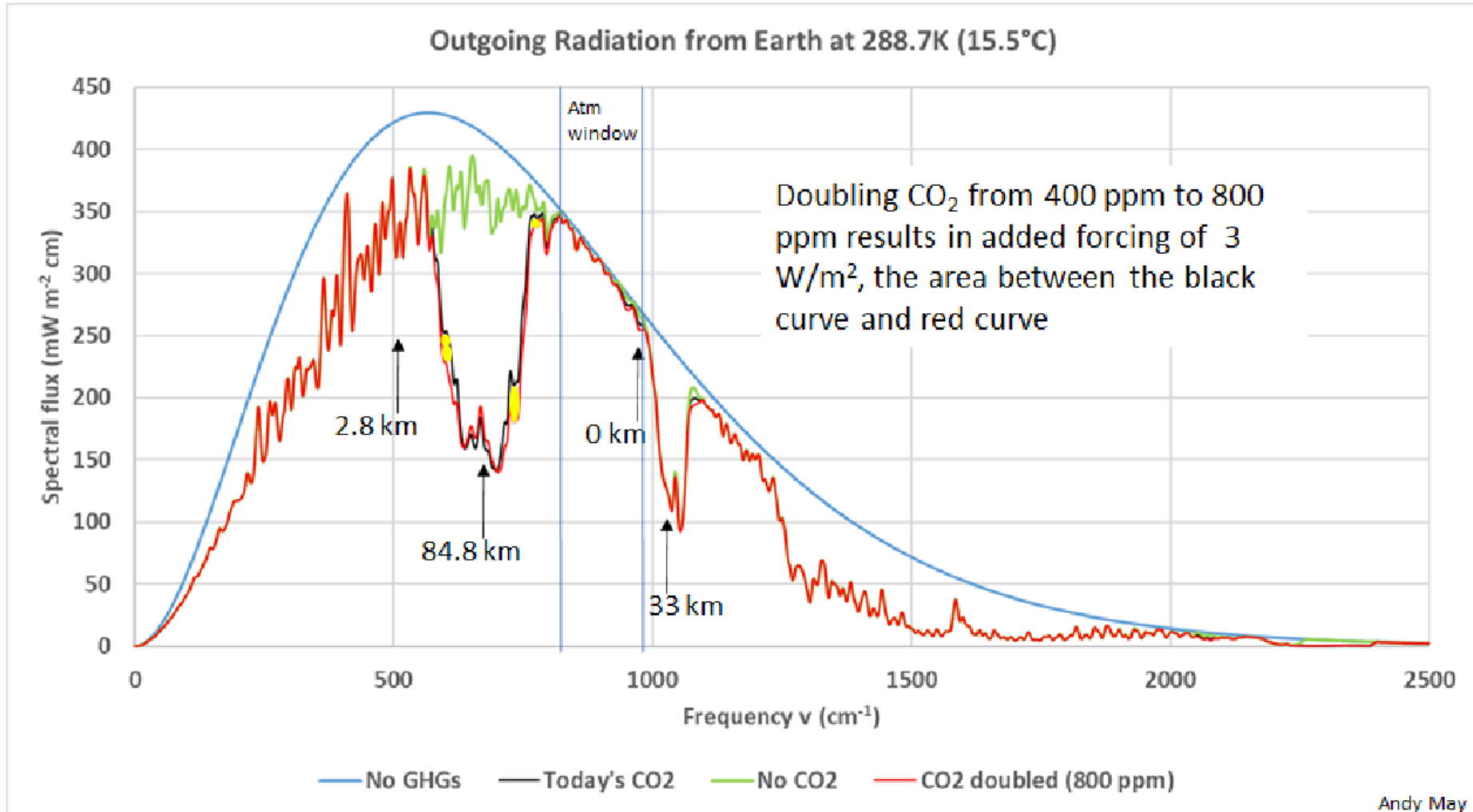
Radiative Forcing – Toay's atmosphere



Radiative Forcing – Which gas plays role ?



Where does it fits in the bigger picture ?



$$\text{Radiative Forcing } (\Delta P) = 5.35 (\pm 0.5) \ln(C/C_0) \text{ W/m}^2$$

Observations

Observations on what is happening with earth systems are important to understand and build evidence

- Before the curve In 1938, Guy Callendar, a steam engineer measuring CO₂ as a hobby compared readings from Kew around 1900 (274 ppm) with readings from the eastern United States in the 1930s (310 ppm). He concluded the atmosphere was accumulating carbon, and that burning coal was the reason. The scientific community did not accept it.
- Twenty years later, Charles Keeling solved that problem by building a record that was Continuous measurement at Mauna Loa and in Antarctica, from March 1958.



Mauna Loa Observatory in the Pacific

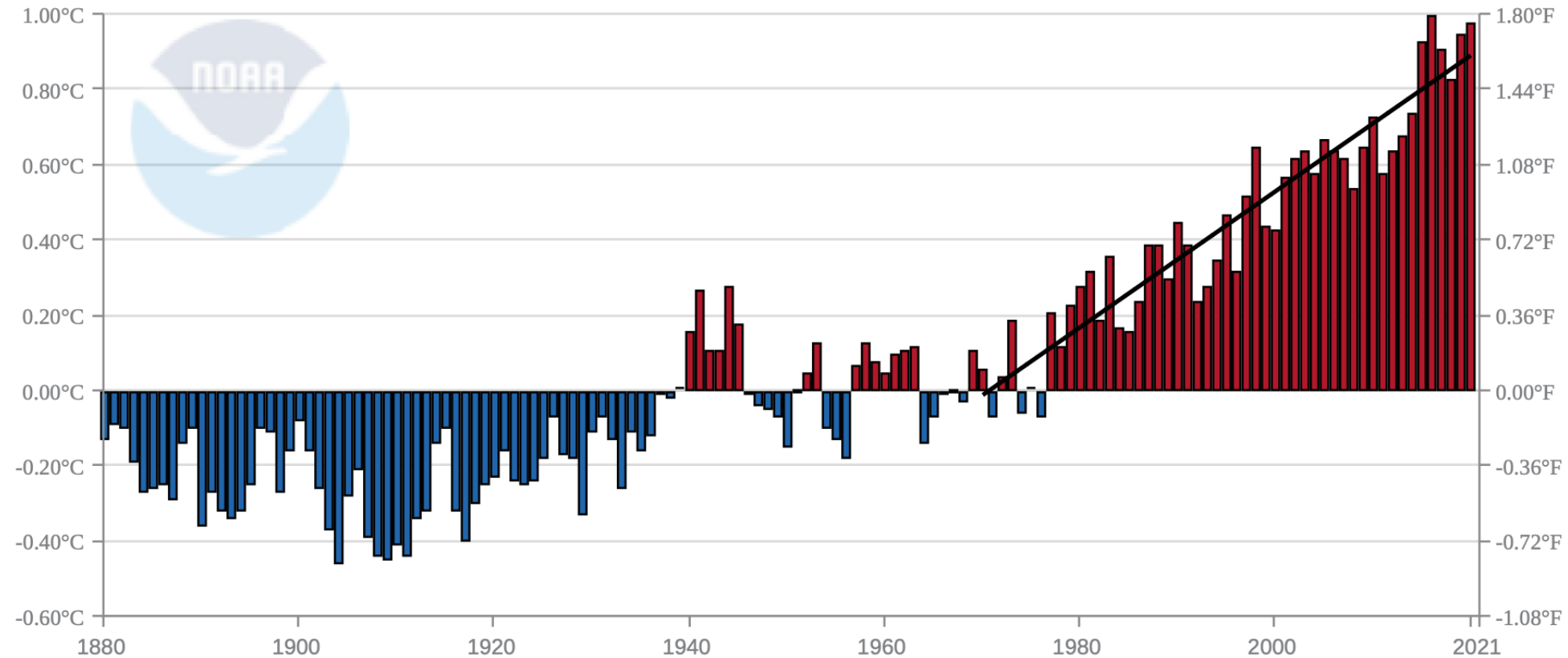
Let's explore Keeling curve:
<https://keelingcurve.ucsd.edu/>

Near Surface Temperatures

Global Land and Ocean

January-December Temperature Anomalies

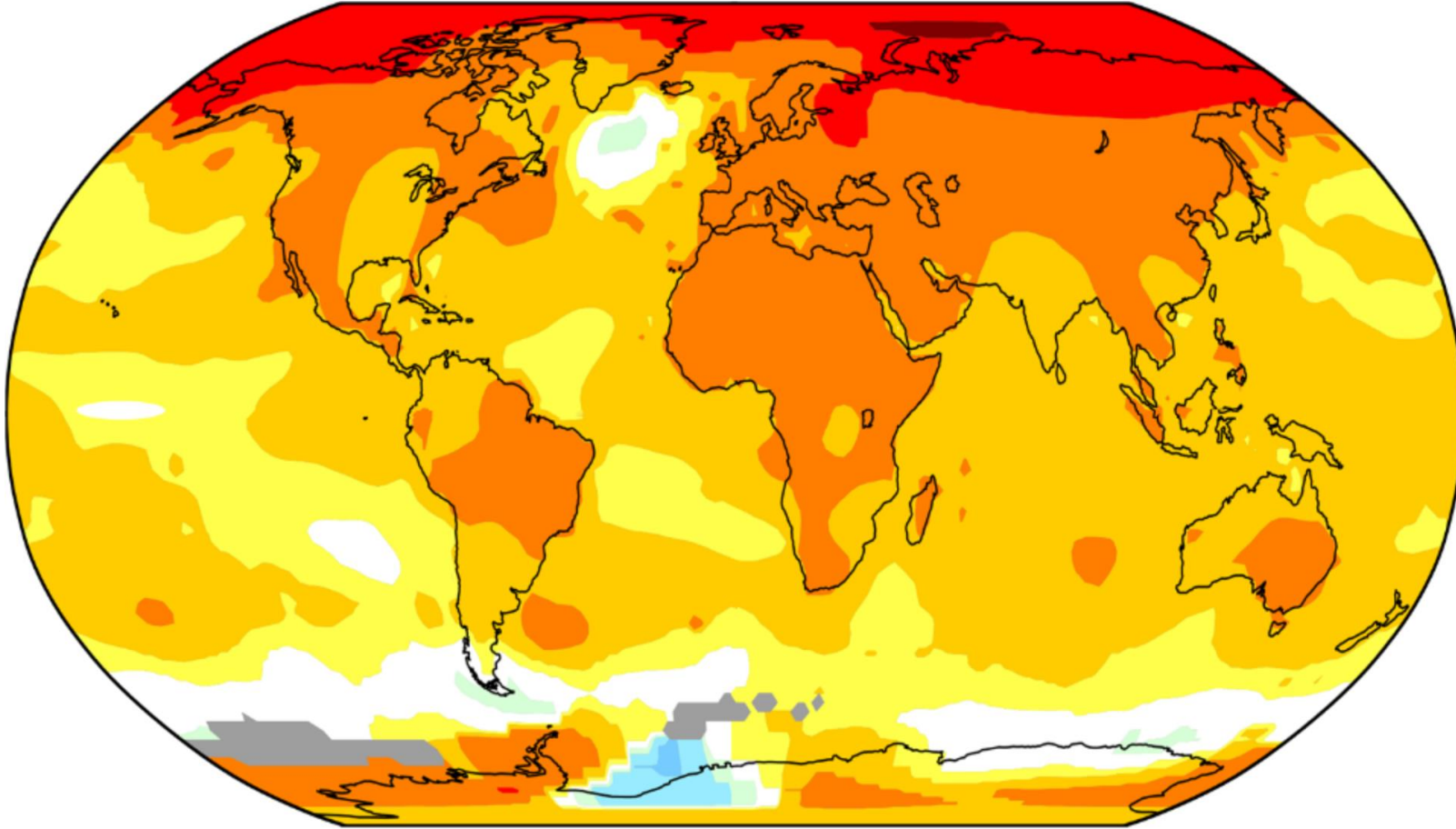
■ 1970-2020 Trend
(+0.18°C/Decade)



National Oceanic and Atmospheric Administration

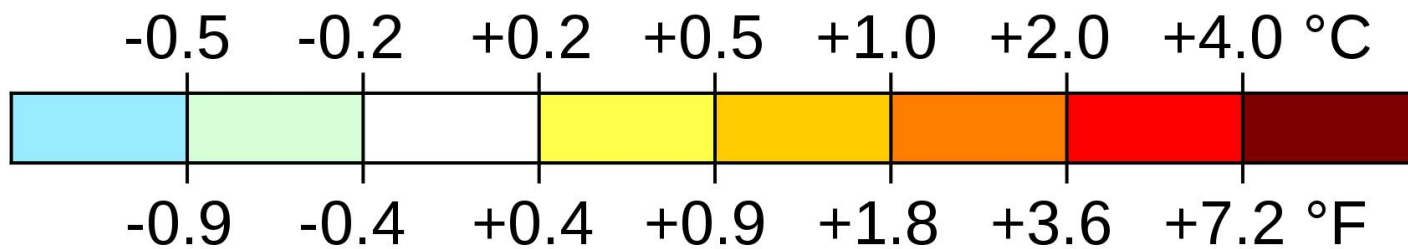
Anomalies are with respect to the 20th century average.

Temperature change in the last 50 years



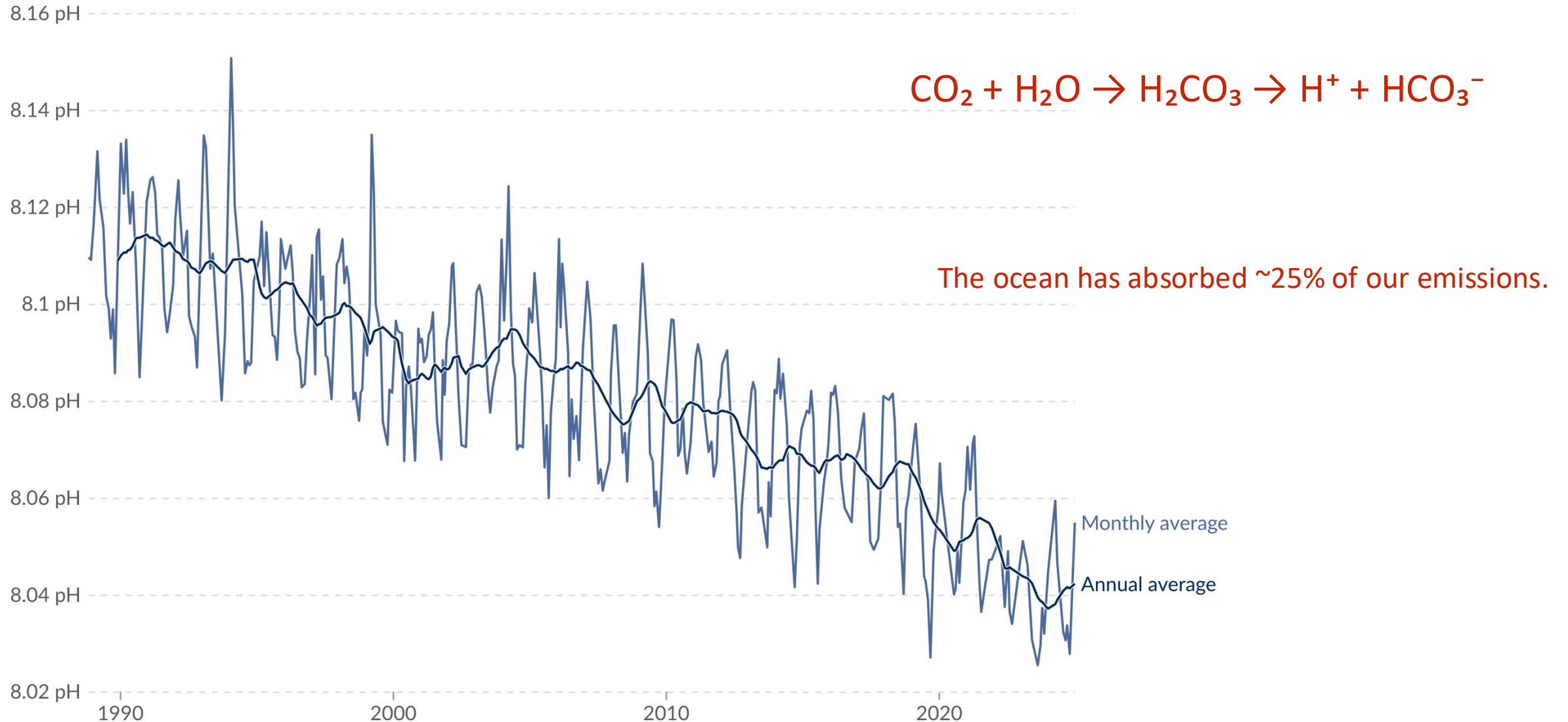
- warming almost everywhere (except in the North Atlantic)
- more warming over land than over the ocean
- largest warming in the Arctic

2011-2020 average vs 1951-1980 baseline



Ocean acidification: mean seawater pH Hawaii

Mean seawater pH is shown based on in-situ measurements of pH from the Aloha station in Hawaii.



Arctic Sea Ice

September 1979



September 2020



What do you think?

Does melting of sea ice contribute to sea level rise?

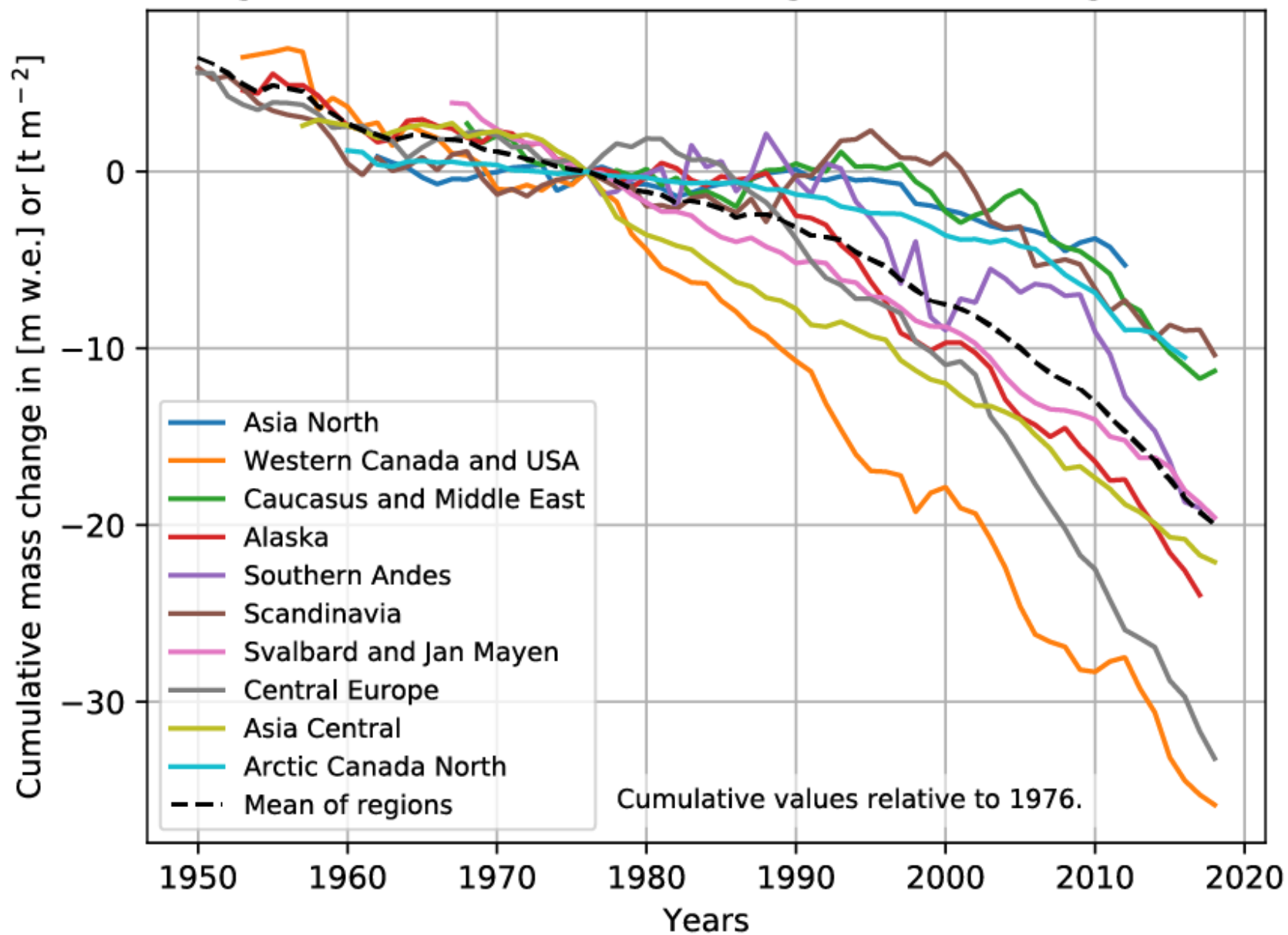


glaciers shrinking worldwide

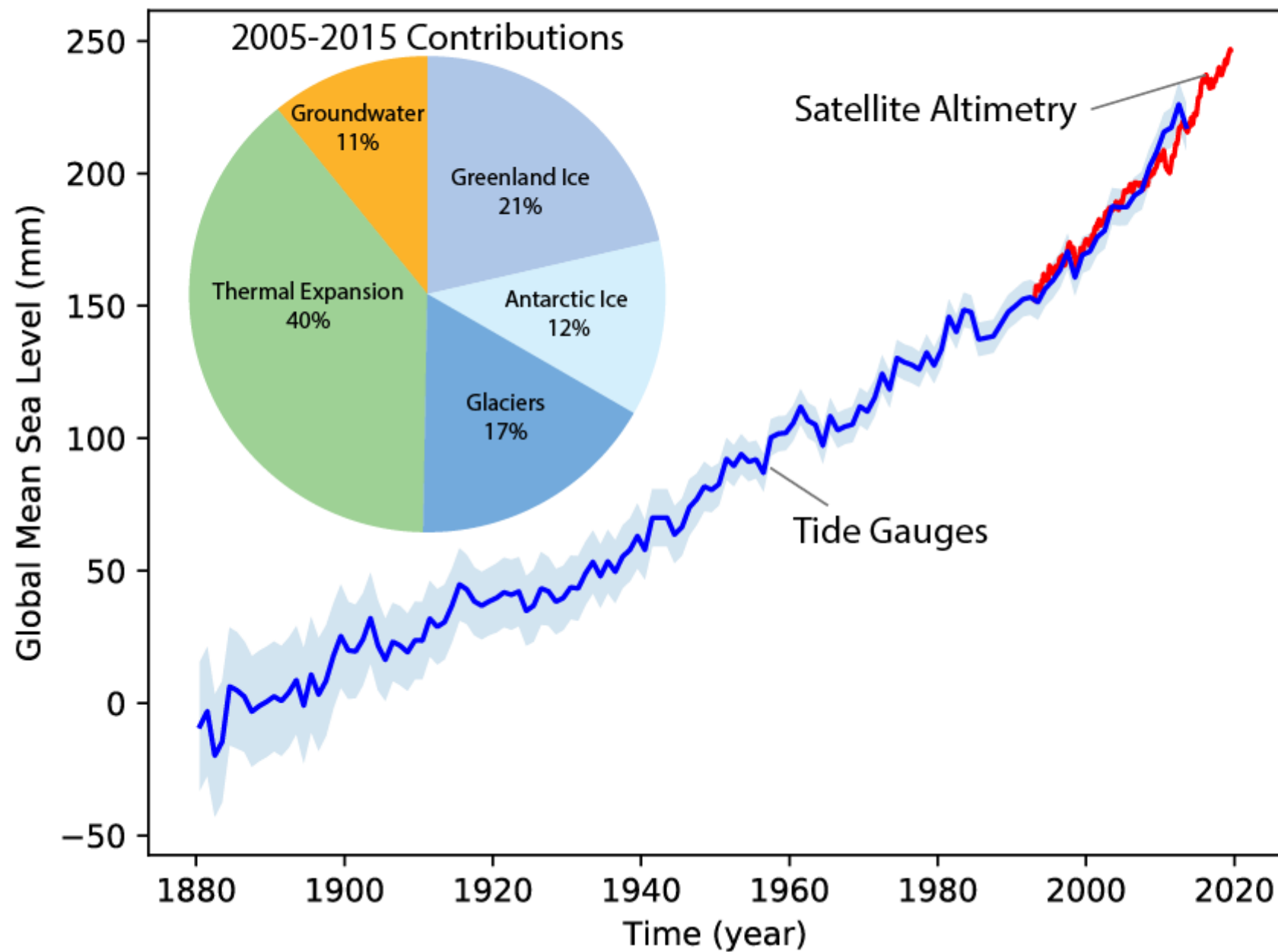


Muir Glacier, SE Alaska
USGS & NPS photographs





Global Mean Sea Level Change



Does humans cause climate change ?

- How climate deniers argue:

<https://www.youtube.com/watch?v=3R7RBtz6-Zk>

- How scientists respond:

https://www.youtube.com/watch?v=J1KGnCj_cfM